

Vivek Moorkoth

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EDUCATION

University of California, San Diego

Expected Graduation June 2029

B.S. Mathematics-Computer Science | GPA: 3.90

- Relevant Coursework: MATH 18, MATH 20C (currently enrolled), CSE 11, CSE 12 (currently enrolled), SIO 25 (Climate Change and Society)
- Other Coursework: COMSC 140 (Python Programming), CIS 22A (Beginning Programming Methodologies in C++), ENGR 100 (Intro to Engineering), AP CSA, AP Calc AB/BC, AP Stats, AP Physics C: Mechanics, AP Chem, AP Bio, AP Microecon, AP Lang, AP US Gov & Politics

RESEARCH & PROJECT EXPERIENCE

Project Team Member – TubeScope, DS3 (Data Science Student Society)

Oct 2025 – Dec 2025

- Built automated Python pipelines to collect, clean, and store large-scale time-indexed data from public APIs.
- Processed and normalized temporal datasets using Pandas, handling missing values and inconsistent reporting intervals.
- Analyzed time-series patterns such as peaks, decay, and persistence to distinguish short-lived vs sustained outcomes.
- Used early-stage temporal signals to infer future behavior, emphasizing interpretability over black-box prediction.
- Created visual summaries and metrics to support data-driven operational decision making.
- Collaborated using Git/GitHub in a team-based, iterative research workflow.

Skills: Python, Pandas, Time-Series Pattern Analysis, EDA, Missing Data Handling, Git/GitHub

Finance/CS Intro Researcher - ASDRP

June 2024 – Aug 2024

- Conducted applied research on real-world datasets characterized by noise, sparsity, and uncertainty.
- Applied statistical and machine learning techniques to infer trends and directional outcomes.
- Evaluated trade-offs between predictive performance and interpretability.
- Participated in weekly research discussions focused on methodology, assumptions, and limitations.

WORK & LEADERSHIP EXPERIENCE

SAT Tutor - Schoolhouse

Oct 2024 – Dec 2024

- Served as an SAT Tutor for Schoolhouse, a global nonprofit platform providing free personalized academic support
- Prepared structured lesson plans and slide decks to teach Math and Reading & Writing sections of the SAT
- Facilitated interactive class discussions and provided targeted instruction for groups of up to 10 students in a virtual learning environment (Zoom)
- Utilized Khan Academy resources to support curriculum and reinforce key concepts
- Adapted teaching strategies to accommodate diverse learning levels and ensure effective student engagement and comprehension

Junior Achievement Intern - JA Norcal

June 2024 – Aug 2024

- Served as Project Leader collaborating with executives from CNBC to develop and pitch a 3-part video series for CNBC Make It
- Managed mentor communications, coordinated group meetings, and led the final project pitch presentation
- Participated in job shadowing experiences at leading companies including Accenture, Goodby, Silverstein & Partners, HSBC, and Cisco
- Awarded First Place in Accenture's Social Innovation Challenge for presenting a high-impact solution

LEADERSHIP EXPERIENCE

REC Foundation Online Challenges: Reverse Engineering Lead - VRC

June 2023 – May 2024

- Led structured analysis and documentation of complex systems by decomposing components and functional relationships.
- Designed organizational frameworks to improve clarity and reproducibility of technical findings.
- Contributed to a submission selected as Global Finalist (Top 5 teams worldwide).

Irish Guard Saxophone Section Leader - Dublin Irish Guard Marching Band & Color Guard

July 2023 – May 2025

- Provided musical and choreographic instruction to a section of 15–20 saxophonists during the marching season
- Organized and facilitated skill-building workshops and team-bonding events to strengthen section cohesion
- Coordinated with instructional staff through pre-rehearsal leadership meetings to ensure efficient rehearsal execution
- Collaborated with other section leaders to design and implement integrated music and visual workshops
- Partnered with adult staff to maintain structured, high-quality rehearsals and overall program effectiveness

SKILLS

Data & Analytics: Python, Pandas, Time-Series Pattern Analysis, Exploratory Data Analysis, Statistical Modeling, Interpretable Decision Support

Tools: Spreadsheet Software (Excel, Google Sheets), Git/GitHub, MATLAB

Programming: Python, Java, C++, JavaScript, HTML/CSS

AWARDS AND HONORS

Circle Scholars \$5,000 Scholarship | Junior Achievement · March 2025

National Merit Commended Scholar | NMSC · May 2024

AP Scholar with Distinction | CollegeBoard · July 2025